

GÖKALP EKER

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EDUCATION

Çankaya University — Ankara, Turkey

09/2023 – Present

B.S., Computer Engineering (3rd Year Student) | CGPA: 2.95/4.00

EXPERIENCE

Energy Market Regulatory Authority (EPDK) — IT Department, Intern

Aug 2026 – Sep 2026

Ankara, Turkey

- Designed and built an IT Assets Management System (IAMS) from scratch — inventory and fault/service-history tracking with role-based access — and presented it to the department.
- Integrated IAMS with Active Directory authentication, automated email notifications, and Zabbix monitoring; deployed to production using Docker and IIS.
- Gained hands-on experience with core networking (protocols, CAT5/CAT6 cabling, RJ45 termination), virtualization/hypervisor technologies, and container platforms (Docker, Kubernetes, OpenShift, ESXi).
- Installed a new physical server in the server room and configured remote access; delivered internal presentations on networking, virtualization, offensive security, and system monitoring (Zabbix, Grafana).

TECHNICAL SKILLS

Languages: C++, C, Python, Rust, Bash

Systems & Infrastructure: Networking, Virtualization, Docker, Kubernetes, OpenShift, ESXi, Zabbix, Grafana, Active Directory, IIS, Linux

Tools & Libraries: Git, GitHub, OpenCV, NumPy, Pandas, PyTorch, Matplotlib, Nmap, Wireshark

Hardware/IoT: ESP32, Arduino, Raspberry Pi, STM32, CC1101, PlatformIO, Blynk

Core Knowledge: Embedded Systems, Computer Vision, Cyber Security, Deep Learning

TECHNICAL PROJECTS

CindraNet — Decentralized P2P Messaging Platform | Rust, Tauri *(Private repository — code available upon request)*

- Built an end-to-end encrypted P2P messaging platform in Rust using X3DH and Double Ratchet, with persistent ratchet state, forward secrecy, skipped-message-key handling, and encrypted local key storage.
- Designed a Kademlia-style DHT for public-key-based peer discovery and a multi-hop onion-routing/relay layer (X25519 layered encryption) with trust/latency-based route selection and NAT traversal via rendezvous relaying.
- Implemented signed, hash-chained group membership state, encrypted redundant backups, cryptographic decommissioning, and peer-level token-bucket rate limiting.
- Built a Tauri desktop client sharing the same Rust protocol core, and developed concurrency/protocol robustness tests (fuzzing-lite, serialization round-trips, state-machine validation), fixing real-world race conditions and mutex poisoning found during multi-node testing.

AI-Generated Image Detection System (ResNet18)

- Developed a binary classification model to distinguish real vs. AI-generated images using transfer learning on the ResNet18 architecture.
- Trained on the CIFAKE dataset (100k+ images) and achieved 97.2% test accuracy on held-out data.

IoT-Based Smart Air Quality & Environment Monitor

- Designed a real-time monitoring system using ESP32 and C++ with integrated DHT22 and MQ-135 sensors.
- Configured local visualization via I2C on SSD1306 OLED and remote monitoring via the Blynk IoT Platform.

Smart Document Scanner

- Developed a C++ & OpenCV pipeline using Canny edge detection and transformation matrices for document warping.
- Enhanced text readability and removed environmental shadows through adaptive thresholding techniques.

LANGUAGES & INTERESTS

Languages: Turkish: Native, English: Advanced

Interests: Edge AI, Open Source Contribution, Generative AI, Distributed Systems, Cryptography & Network Security